Mian Zhang

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Education

The University of Texas at Dallas — 2024–Spring 2027 (expected)
Ph.D. in Computer Science advised by Zhiyu Zoey Chen — *Dallas, TX*

Soochow University — 2020–2023
M.S. in Computer Science advised by Wenliang Chen — *Suzhou, China*
- Outstanding Graduate

Research Interests

LLM & Agent Evaluation · Reinforcement Learning · Reliability · Efficiency

Research Experience

Scale AI — May. 2026–Aug. 2026
Research Intern — *San Francisco, CA*
Hosts: Manasi Sharma, Ying Liu, Daniel Yue Zhang
- **Multi-Agent Computer Use**: Identified the non-mergeability of VM states as a critical bottleneck in multi-agent computer use; developed Spine-Branch Coordination [1] to preserve state continuity while enabling parallel task execution.
- **GUI-Agent Evaluation**: Evaluated Spine-Branch on long-horizon tasks, improving task success rates by 6.0–16.5 percentage points while reducing per-task costs by 34–70%.

Zoom GenAI — May. 2025–Aug. 2025
Research Intern — *Seattle, WA*
Hosts: Kaiqiang Song, Zheng Chen
- **Synthetic Data & Evaluation**: Developed LogicIFGen [1], a scalable framework for synthesizing complex, verifiable instructions with executable ground truth and intermediate-state tracking. Constructed LogicIFEval [1], a benchmark of 426 logic-intensive tasks.
- **Reinforcement Learning**: Constructed LogicIFTrain [1] with 27K verifiable training examples and applied RLVR to improve logic-following accuracy by 10.7 percentage points, with generalization to mathematical reasoning, coding, and other tasks.
- **Tool-use Agent Evaluation**: Contributed several realistic, multi-step tool-use scenarios to LiveMCP-101 [6], a benchmark evaluating MCP-enabled agents.

Tencent AI Lab — May. 2024–Aug. 2024
Research Intern — *Seattle, WA*
Hosts: : Linfeng Song, Haitao Mi, Dong Yu
Research on robust preference learning.

The University of Texas at Dallas — Aug. 2024–present
Research Assistant — *Dallas, TX*
Advisor: Zhiyu Zoey Chen
- **Agentic RL (for Search Agent)**: Research on efficient agentic search; designed metrics to quantify over-search and under-search behaviors in LLM agents [5] and developed HiPRAG [3], improving accuracy by 5.1 points while reducing search calls by 29%.
- **Multi-Instruction RL**: Identified exploration bias in RL when multiple instruciton coexist [4]; proposed metrics and a mitigation methods for RL training [4], improving instruction-following accuracy by 4.5 points across three benchmarks.
- **LLM Evaluation (Health)**: Developed a hierarchical benchmark CBT-Bench [9] and a preference dataset PsyCoPref [8] for evaluating and improving LLMs in mental health applications.

Tencent 2022
Research Intern *Shenzhen, China*
Mentor: : Lifeng Jin
Research on dialogue understanding, inconsistency, and safety [9].

Papers (Selected)

The full publication list is at Google Scholar. * denotes equal contribution.

[1] **Spine-Branch Coordination for Multi-agent Computer Use** Preprint
Mian Zhang, Manasi Sharma, Sheng Zhang, Minlai Yang, Kejian Shi, Ying Liu, Zhiyu Zoey Chen, Daniel Yue Zhang

[2] **LogicIF: Towards Complex Logic Instruction Following** COLM 2026
Mian Zhang, Shujian Liu, Sixun Dong, Ming Yin, Yebowen Hu, Xun Wang, Steven Ma, Song Wang, Sathish Reddy Indurthi, Haoyun Deng, Zhiyu Zoey Chen, Kaiqiang Song

[3] **HiPRAG: Hierarchical Process Rewards for Efficient Agentic Retrieval Augmented Generation** ICLR 2026
Peilin Wu, **Mian Zhang**, Kun Wan, Wentian Zhao, Kaiyu He, Xinya Du, Zhiyu Chen

[4] **Mitigating Exploration Bias in RL for Multi-Instruction Following** EMNLP 2026
Mian Zhang, Yueqin Yin, Kaiyu He, Peilin Wu, Xinlu Zhang, Mingyuan Zhou, Zhiyu Zoey Chen

[5] **Search Wisely: Mitigating Sub-optimal Agentic Searches By Reducing Uncertainty** EMNLP 2025
Mian Zhang*, Peilin Wu*, Xinlu Zhang, Xinya Du, Zhiyu Zoey Chen

[6] **LiveMCP-101: Stress Testing and Diagnosing MCP-enabled Agents on Challenging Queries** Preprint
Ming Yin, Dinghan Shen, Silei Xu, Jianbing Han, Sixun Dong, **Mian Zhang**, Yebowen Hu, and etc.

[7] **MMTok: Multimodal Coverage Maximization for Efficient Inference of VLMs** ICLR 2026
Sixun Dong, Juhua Hu, **Mian Zhang**, Ming Yin, Yanjie Fu, Qi Qian

[8] **Preference Learning Unlocks LLMs' Psycho-Counseling Skills** ACL 2026
Mian Zhang, Shaun M. Eack, Zhiyu Zoey Chen

[9] **CBT-Bench: Evaluating Large Language Models on Assisting Cognitive Behavior Therapy** NAACL 2025
Mian Zhang, Xianjun Yang, Xinlu Zhang, Travis Labrum, Jamie C. Chiu, Shaun M. Eack, Fei Fang, William Yang Wang, Zhiyu Zoey Chen

[10] **SafeConv: Explaining and Correcting Conversational Unsafe Behavior** ACL 2023 Oral
Mian Zhang, Lifeng Jin, Linfeng Song, Haitao Mi, Wenliang Chen, Dong Yu

Selected Awards

Tinker Research Grant from Thinking Machines	2025
Researcher Access Program from OpenAI	2025
Outstanding Graduate Award at Soochow University	2023
First-class Scholarship at Soochow University	2020–2021
Principal's Award	2019
Mathematical Contest in Modeling National First Prize (top 1%)	2018